

ATHARVA- An Efficient Framework for Ocean Acoustic Data Management

Parvathy Ramakrishnan P, Sahely Bhadra, Sudina Dinesh, Satyajit Das

(142203002@smail.iitpkd.ac.in, sahely@iitpkd.ac.in, sudhinadinesh@gmail.com, satyajitdas@iitpkd.ac.in)

Department of Data Science, Indian Institute of Technology Palakkad, Kerala, India

Abstract—In today’s data-centric era, efficient data management is paramount due to the sheer volume of data generated. This is particularly crucial in ocean observation systems, where extensive data is produced through advanced platforms and sensors. Sonar data, vital for underwater acoustic modeling, poses challenges in data integration and analysis due to diverse packet formats and decoding overhead for bytestream data packets. Enabling data scientists and other analysts to find and access relevant data from the data collected can also be a challenge, especially when the data is spread across various big data systems. To address these issues, we propose a framework for ocean acoustic data management, Acoustic daTa Handling ARchival Visualization and Analytics (ATHARVA). Our framework includes 2 key components (i) A content based indexing for fast and efficient data discovery with minimum overhead and (ii) A GUI-based software tool for interactive information retrieval and visual analysis of data. The proposed framework ATHARVA ensures seamless integration of diverse data, reducing manual efforts and modernizing data management. Demonstrating efficiency in data discovery, analysis, retrieval, and visualization, ATHARVA promises a significant impact on ocean data management.

Index Terms—Data Management, Sonar, Byte stream, Data Integration, Content Based Indexing, Data Visualization

I. INTRODUCTION

Data is being created and consumed at unprecedented rates, and data management solutions are essential for making sense of the vast quantities of data. Data management is an umbrella term that includes multiple facets such as data quality checks, access control, integration, database design, and the pivotal elements of storage and archiving. It connects the data acquisition components to the modeling and analysis components.

One of the domains where this is especially relevant is ocean observation systems [1], [2], where vast amounts of data are being generated through advancements in ocean observation platforms, sensors, and sonar technologies. These data over the oceans are complex and time series observations are vital. The data collected over prolonged periods is extremely useful for scientific analysis. Proper methods to collect, organize and analyze these data will increase the usability and efficiency of data.

Ocean data management has many open challenges [3]. They range from storing and efficiently retrieving the enormous volume of data collected to building software frameworks for data processing and visualization. In this paper, we propose a framework for ocean acoustic data management, addressing the challenges in storing, retrieving, and processing vast ocean data [8].

Efficient data storage and retrieval is a key aspect of data management, particularly in scenarios with lots of data to manage. The large volume of data generated has to be stored somewhere and must be easily accessible later on for further analysis. While file systems are often chosen for long-term data archival and storage due to their flexibility and scalability, challenges arise in efficient retrieval and managing the substantial volume of data.

File systems can accommodate diverse data types and formats more effectively than databases, which might be optimized for specific data structures. In situations where the primary need is to read or write large chunks of data sequentially (e.g., in data processing workflows), file systems can be more efficient than databases that are optimized for random access. Raw data files are less prone to obsolescence and can be easily transferred between systems or archived for long-term storage without dependencies on specific database technologies.

However the file system data storage faces challenges in efficient data retrieval. While files can be searched based on attributes like name, date, and size, searching through the file system for matches can become cumbersome, especially when users need to identify files based on specific search terms or values within a certain range. Enabling data scientists and other analysts to find and access relevant data can be a challenge, especially when the data is spread across various big data systems.

Data integration and analysis also has difficulties due to the diverse packet formats and decoding overhead for the byte stream data packets. Domain knowledge and manual efforts are needed to understand the packet structure, decode and interpret the packets and to plot the trends and anomalies in data.

To address these issues, we propose a framework for ocean acoustic data management, Acoustic daTa Handling ARchival Visualization and Analytics (ATHARVA). Our framework leverages the idea of content-based indexing to help make data more accessible. In practical terms, indexing files based on variables creates a navigational aid, making it more convenient to locate and access data points relevant to specific criteria. This systematic approach significantly enhances the speed and accuracy of data retrieval, particularly crucial for managing large datasets, thereby saving time and resources. We have built a software tool on top of indexing to enable fast data retrieval and seamless integration of data from various sources, minimizing manual intervention in data analysis and

Modelling & Analysis	Forecast Models	System Monitoring	Decision Support	Dashboards & WebApps	Deep Learning Models
Data Management	Data Integration	Data Quality Checks	Data Access Control	Database Design	Data Storage & Archiving
Observing systems	Sonars	Radars	Moored Buoys	Satellites	Fixed Stations

Fig. 1: The ocean data management system connecting the observing systems to the modeling and analysis components

modernizing ocean data management.

The rest of this paper is organized as follows: Section 2 describes the background and motivation of ATHARVA, outlining the key factors that drove its development. Section 3 details the architecture, methodologies, and components of ATHARVA. Section 4 analyzes and discusses the performance results. Finally, Section 5 serves as the conclusion of the paper, summarizing the key findings.

II. BACKGROUND AND MOTIVATION

Acoustics play a central role in ocean observation. Sound Navigation and Ranging (SONAR) act as the eyes and ears of a naval vessel. The data collected from sonar applications hold immense potential for modeling the underwater environment. These data collected over prolonged periods are extremely useful for scientific analysis and informing policy decisions. The applications are in antisubmarine warfare to detect, track and neutralize enemy ships and submarines.

Sonar array consists of hundreds of sensors that pick up the acoustic signals and convert them to electrical signals which are captured into binary files. During ocean trials, sonar sensor data is recorded using a dedicated data recorder subsystem. Trial-specific environmental data such as latitude, longitude, depth, etc., and transmission-specific data such as pulse type, frequency, GPS speed, etc. are also recorded. All these associated oceanographic and transmission-specific parameters that defines the acoustic data collected are termed metadata. They are critical for understanding and retrieving the acoustic data effectively.

The generic digital data recorder (GDR) receives the data packets without any particular order. Acoustic data packets often have very high-frequency components, while non-acoustic packets may have lower update frequencies, typically less than 1Hz. GDR appends a header with each packet received. Header includes the time-stamp of packet arrival, IP Address, port, sequence number, and packet size. This information is used for packet filtering and replaying the acoustic signals. Each sonar trial data packet to be stored is accompanied by header information. This header has the information needed to filter packets as acoustic and non-acoustic metadata packets.

Managing this complex binary data file with multiple sources, diverse packet structure and packet rates from the file sytem storage where they are dumped after the trials has many challenges. The key challenges are,

- Binary file format- The raw data file generated from sonar is binary encoded and requires additional effort and time to parse data. The packet structure used during encoded is also needed to decode the file into readable units.
- The scale of data generated- The huge volume of data generated has to stored and analyzed.
- Traditional file system storage- Search and discovery of relevant data is difficult especially as data is spread across various systems and is in binary format.
- Diversity in the types of data to be handled- The dataset contains data measured using different sensors , multiple packet structures, variable packet rate & packet size. The metadata parameters collected also vary between different trails. Integrating this complex data and providing a uniform querying interface.
- Need for easy to use software tools- Manually handling the data, maintaining metadata, and analyzing and plotting inferences consumed an enormous amount of time and demanded specialized domain knowledge skills

Motivated from the above challenges, we proposed ATHARVA, a framework for ocean acoustic data management. The proposed framework aims to address the need for an efficient and streamlined method to search and discover relevant data at scale. Modernizing the data management systems using software systems and tools will help in increasing the availability and utility of data. The key design goals in consideration were,

- Design an integrated framework capable of handling diverse data formats, variable packet rates, and sizes.
- Address big data challenges in data storage, retrieval and analysis.
- Ensure data accessibility and usefulness.
- Improve traditional file system storage by reducing retrieval time and enhancing data discovery through tagging and indexing.
- Minimize manual intervention in data analysis by building easy-to-use system tools and interfaces

III. PROPOSED FRAMEWORK

Our proposed framework ATHARVA includes 2 key components, (i) An indexed metadata database for fast and efficient data discovery with minimum overhead and (ii) A GUI-based software tool for interactive information retrieval and visual analysis of data.

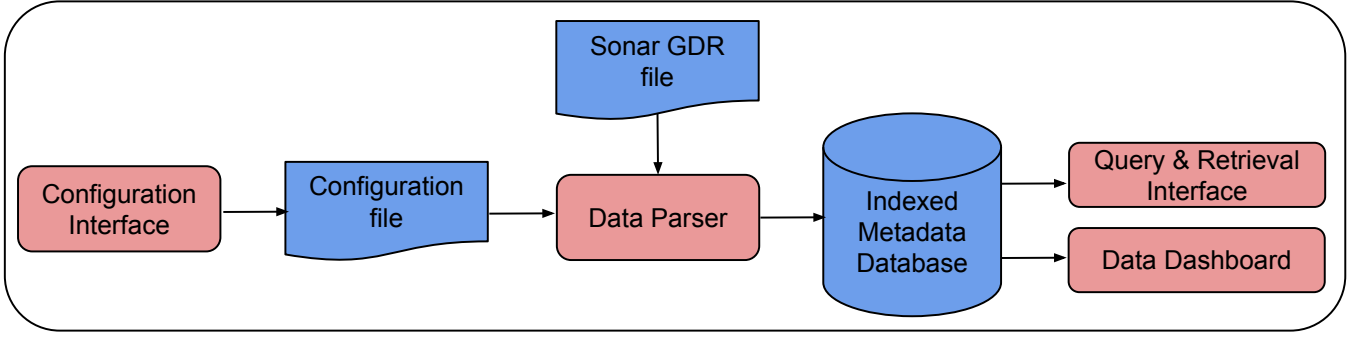


Fig. 2: The data flow within the proposed framework.

The central concept of our framework is to create an indexed metadata database from datafiles, enabling easy discovery of acoustic data based on metadata queries. Additionally, the byte offsets and timestamps are also captured into the indexed metadata database which facilitates pinpointing relevant data within GDR files. This will eliminate the need to read entire files during retrieval. To optimize performance, we employ indexing and partitioning techniques within the database. These strategies reduce the data volume involved in queries, resulting in faster query execution. The software tool is designed to perform the following tasks. (i) Facilitate easy data integration (ii) Process binary-encoded sonar GDR files (iii) Load the metadata extracted from these files into the database (iv) Enable users to search and find the sonar data files for specific information (v) Provide a visual representation of the metadata and acoustic data.

Figure 3 shows the data flow in the proposed framework. The modules configuration interface, data parser, query and retrieval interface and data dashboard are implemented in GUI tool using python pyqt [4]. The indexed metadata database is implemented using PostgreSQL database.

A. Configuration Interface

The key functionality of this interface is data integration. This involves harmonizing data formats, resolving discrepancies, and consolidating information into a unified structure for efficient analysis and visualization. The configuration interface will collect the file structure information required to decode binary file from user through the GUI and will automatically create configuration file for the project

As there are files from different sources with varying packet structure and parameters, a standard way to describe the file structure is needed [5]. A configuration format in JavaScript object notation (JSON) is designed as a standard way to describe sets of recorded digital signal samples during a project. This enables recording all project-specific details in one place and passing it as initial input to the tool for adding a new project. The configuration file is defined in plain text, and structured with JSON which can capture the file structure, packet types, and data expected within each packet. It has 3 components. (i) Ports: Details of the destination port and

packet size expected, source IP, Packet source & Packet type are to be defined here. This data is used for filtering packets from GDR accordingly. (ii) Packets: Defines the packet structure. Includes field names, data type & size (iii) Parameters: Allows the user to choose what all parameters need to be stored and indexed in the database. The configuration file is flexible and allows the data analysis framework to parse files with different structures thus ensuring data integration among multiple file formats

B. Data Parser

The data parser module in the framework is responsible for processing and extracting the metadata from the sonar data files stored in the file system. The data parser along with the configuration file will convert the binary to human-readable engineering units. It is designed to automatically extract metadata from new files added to selected file system folders to facilitate more sophisticated searches. The users can also choose particular files and project it belongs to from the interface to parse files in other folders. The metadata is then stored in the indexed metadata database. This database is used to search and discover relevant data.

The algorithm 1 details the data parsing and automatic metadata extraction. The file is read sequentially using header to filter out acoustic and metadata packets. Byteoffset is incremented on reading each packet. Acoustic packets are skipped during parsing and metadata packets are decoded using project configuration file. The user specified parameters in the configuration file are alone loaded to database. Parameter name and value along with byteoffset(packet position within the file) and timestamp from header is inserted into the database.

C. Indexed Metadata Database

Traditionally, file systems allows for the search and retrieval of files based on filename, size, pathname and so on. However, for our purposes, simply relying on pathname-based retrieval is insufficient, as we also require the ability to find collections of files based on their metadata attributes. Consequently, the standard file system falls short for our needs, as it necessitates exhaustive crawling of the entire file system, which becomes

Algorithm 1 Data Parsing Algorithm

```
1: headersize, acoustic_ports ← json.load(config)
2: db ← MyDatabase()
3: for all file in filestoparse do
4:   bytearray ← 0 and metalist ← empty list
5:   db.file_detail(fname, floc, date, project, fsize)
6:   while True do
7:     Header ← file.read(headersize)
8:     bytearray ← bytearray + headersize
9:     if not Header then
10:      file.close()
11:      break
12:     end if
13:     portnum, packetsize, sourceIP, timestamp ← np.frombuffer(Header, metaheader)
14:     if portnum in acoustic_ports then
15:       file.seek(packetsize)
16:       bytearray ← bytearray + packetsize
17:     else
18:       packettype ← json.load(config)
19:       prmlist ← parameters[packettype]
20:       data ← file.read(packetsize)
21:       bytearray ← bytearray + packetsize
22:       metapacket ← np.frombuffer(data, packettype)
23:       for all prm in prmlist do
24:         metalist.append(prm, value, bytearray, timestamp)
25:       end for
26:     end if
27:   end while
28:   db.insert(metalist)
29:   file.close()
30: end for
```

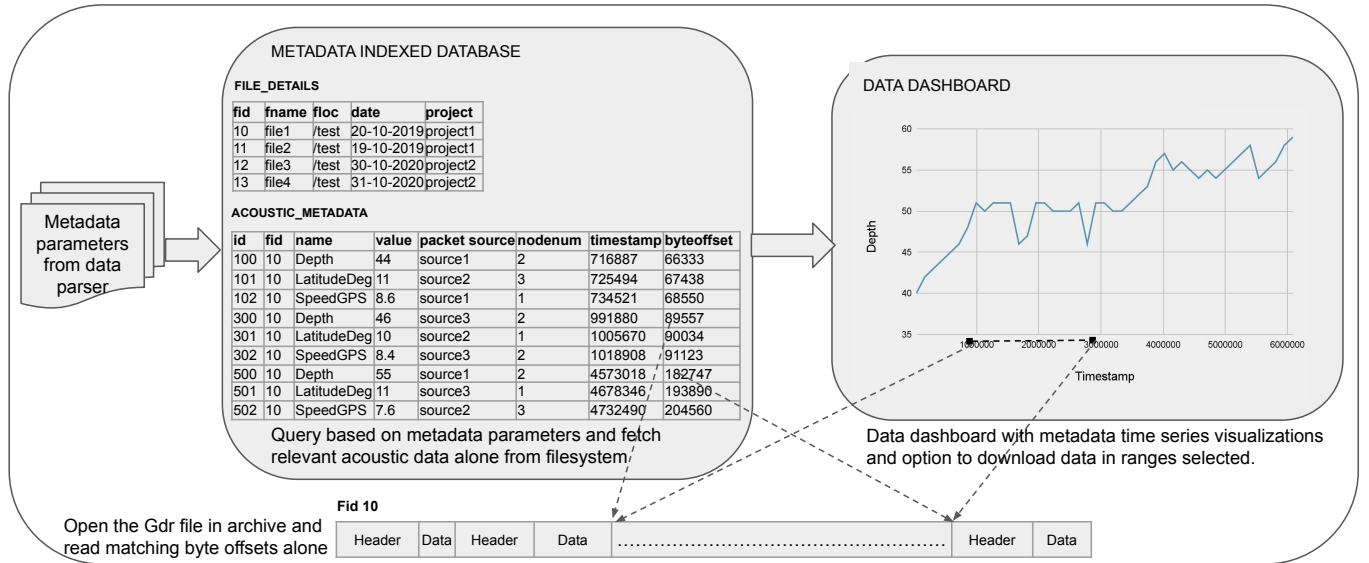


Fig. 3: The overview of the metadata indexed database showcasing the data retrieval and data visualization components build on top of database. The byte offset captured in database helps in streamlined data discovery and access within files.

prohibitively costly as the scale of data grows. In our data management framework, we employ an indexed metadata database utilizing inverted indexing [6] to efficiently store and index metadata packets associated with sonar byte stream files. This approach draws inspiration from the widely used inverted metadata index structure utilized in internet searches [7] to identify pages containing specific search terms.

By adopting an indexed metadata database, our solution offers significant advantages:

- **Customized Search Queries:** Our metadata indexing enables search queries based on both user-defined metadata attributes and system-defined attributes, such as standard file system stat attributes. This flexibility allows users to search for files based on a wide range of criteria.
- **Efficient Retrieval:** By avoiding the need to crawl the entire file system, our approach drastically improves retrieval efficiency. Instead of scanning through every file, the index allows for targeted searches based on metadata attributes, leading to faster and more accurate results.

Fig 4 shows the overview of metadata indexed database. The metadata packets from sonar file is only captured into database. The acoustic data is not stored as this needs large storage and will cause data duplication as actual files are not removed post-processing. The inverted indexing is implemented in PostgreSQL using two tables. An entry for each file is added to the file details table which stores details of the file like name, location, date, and project name, and a unique key, fid is generated. While parsing the file the metadata packets are captured and loaded into the acoustic metadata table. The acoustic metadata table stores the parameter name, value, node number, packet source, timestamp, and byte offset. Timestamp is retrieved from header and byte offset is calculated during the parsing. It indicates the position of the packet within the file and will help in the efficient retrieval of the associated acoustic data precisely. A Btree index is built over the column's parameter name and value for fast data retrieval.

D. Query & Retrieval Interface and Data Dashbaord

The query & retrieval interface and the data dashboard allow users to search relevant data, analyze the data, understand trends and anomalies, and download relevant acoustic data or metadata alone for further analysis. A querying interface is available in GUI using drop downs to choose parameters, operators, and values. The interface will convert user inputs into SQL queries and retrieve files matching all the query conditions. There are options to download the associated acoustic data and metadata for further analysis.

The metadata operations access the database only while the acoustic data operations will query the matching files and the byte offsets within the file from the database and then read the file for the relevant data. The query management and retrieval interface allow users to query acoustic data based on metadata parameters stored in the database. The data dashboard interface is designed to help users get inferences and insights from data. Map-based visualization is available from which users can directly select data of the interested region. Time series

visualization for metadata parameters is also available which will make it easier to identify patterns, trends, and outliers. Acoustic data visualizations includes sensor wise time series overlays, sensor wise mean and variance stem plots, heat maps and power spectral density plot.



Fig. 4: Query and retrieval interface. The query widget has multiple search criterias. Metadata and acoustic data plot widgets and parameters and types of plots needed can be choosed using buttons and dropdowns provided.

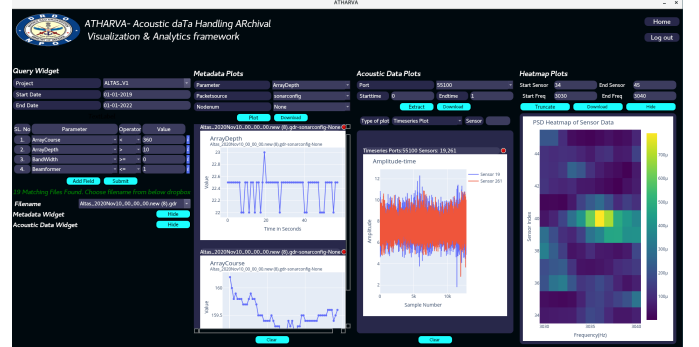


Fig. 5: Query and retrieval interface with metadata plots and acoustic data plots. Different types of visualizations are added for acoustic data including heatmaps, timeseries overlays, power spectral density plots and stemplots

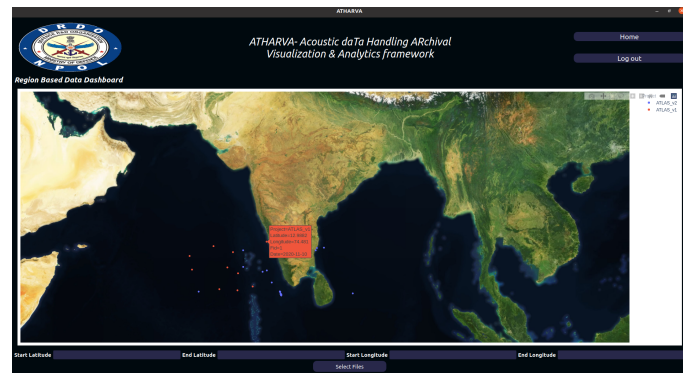


Fig. 6: Data Dashboard with map based region wise data distribution.

IV. RESULTS/DISCUSSION

In this section we discuss the experiments performed showcase the efficacy and performance of ATHARVA. We have used 500GB of sonar trial dataset collected from the Naval Physical and Oceanographic Laboratory for the experiments. This data contains high-frequency ocean acoustic observational data along with oceanographic and transmission-specific parameters. The four key aspects used for evaluation are data parsing speed, data storage overhead, response time and scalability.

A. Data parsing speed

- The data parsing speed is calculated as the total file size parsed divided by the total time taken. ATHARVA has achieved high-speed data parsing and metadata capture, with the capability to process data at a remarkable speed of 6GB per minute. Furthermore, we have automated the data parsing and metadata capture. Once new files are added in the file system folders it will automatically be parsed and indexed into the database.

B. Data storage overhead

- In this experiment the database size is recorded after parsing the dataset. Our database design achieves fast data discovery within the file system with a minimal overhead of 1GB of database over 500 GB of GDR files. This is achieved by storing and indexing only the user specified attributes along with their timestamp and byte offset into database.

C. Response Time

- In this experiment, the average response time for major tasks performed using tool are recorded. Table 1 shows the average response time for all tasks. The metadata operations are in order of milliseconds and acoustic data operations are in seconds. This is because metadata is fetched directly from the indexed metadata database while acoustic data needs to read the GDR file also. Also, the volume of data handled is very huge in the case of acoustic data hence it is processed and visualized in blocks of 60 seconds. Table 2 shows the sample use cases for querying files, the query generated by tool in the back end and the run time taken to execute the query and list files with matching data.

Task	Response Time (sec)
Parse 1GB GDR file	10.0
Retrieve and Visualize metadata	0.59
Download metadata	0.18
Retrieve acoustic data	3.5
Visualize Acoustic data	0.81
Download acoustic data	8.0

TABLE I: Response time for various tasks using tool

D. Scalability

- In this experiment, the use case Q6 in the table 2 is executed across dataset with varying total file sizes. Figure 7 illustrates the relationship between the total file size and query

time. As indicated by the data points on the chart, the query time increases gradually as the size of the dataset (measured in gigabytes) grows. However, the rate at which the query time increases is logarithmic rather than linear. The logarithmic time complexity implies that the search time increases at a slower logarithmic rate as the dataset expands. In practical terms, even as the volume of data grows, the additional time required for a search operation remains significantly less compared to what one would experience with a linear search. This logarithmic behavior in query time complexity is a result of employing an index on the relevant columns. Instead of scanning the entire dataset sequentially, the database engine adopts a "divide and conquer" approach through the indexed structure. This highlights the advantage of employing indexing in scenarios where rapid and precise data retrieval is crucial.

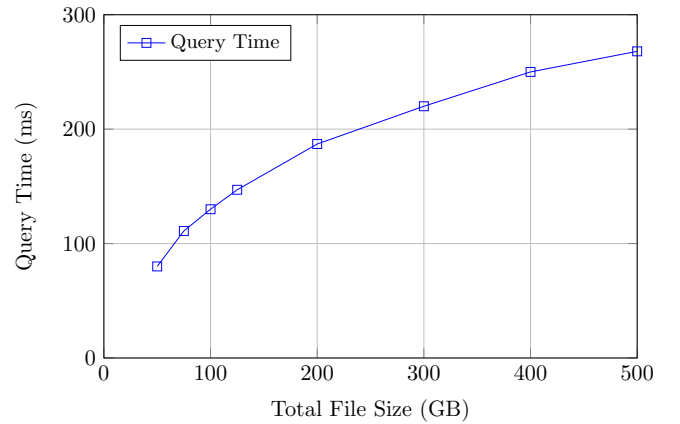


Fig. 7: Query Time vs. Total File Size

V. CONCLUSION

In this work, we introduced an integrated framework, ATHARVA for handling ocean acoustic data ensuring data accessibility and its potential for reuse. The framework is capable of integrating data from different sources having diverse data formats, variable packet rates, and sizes. It has minimized manual intervention in data analysis and modernized the data management system. We proposed an indexed metadata database as a part of the framework that enhances data discovery through tagging and indexing metadata, ensuring data discovery with reduced data retrieval times. The experimental results demonstrate that the framework performs data discovery, analysis, retrieval, and visualization efficiently.

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Sl.No	Search Criteria	Query Executed	Run Time
Q1	Project	SELECT FNAME FROM FILE_DETAIL WHERE PROJECT='ATLAS_v2'	00:00:00.032
Q2	Date	SELECT FNAME FROM FILE_DETAIL WHERE date(date) >'2020-01-01' AND date(date) <'2020-12-01'	00:00:00.047
Q3	Project+Date	SELECT FNAME FROM FILE_DETAIL WHERE PROJECT='ATLAS_v2' AND date(date) >'2020-01-01' AND date(date) <'2020-12-01'	00:00:00.035
Q4	Project+Date+ Parameter1 (Depth)	SELECT FNAME FROM FILE_DETAIL WHERE PROJECT='ATLAS_v2' AND date(date)>'2020-01-01' AND date(date)<'2020-12-01' AND FID IN (SELECT DISTINCT FID FROM ACOUSTIC_METADATA WHERE NAME= 'ArrayDepth' AND VALUE <100 INTERSECT SELECT DISTINCT FID FROM ACOUSTIC_METADATA WHERE NAME= 'ArrayDepth' AND VALUE >50)	00:00:00.205
Q5	Project+Date+ Parameter1 (Depth) +Parameter2 (SpeedLog)	SELECT FNAME FROM FILE_DETAIL WHERE PROJECT='ATLAS_v2' AND date(date)>'2020-01-01' AND date(date)<'2020-12-01' AND FID IN (SELECT DISTINCT FID FROM ACOUSTIC_METADATA WHERE NAME= 'ArrayDepth' AND VALUE <100 INTERSECT SELECT DISTINCT FID FROM ACOUSTIC_METADATA WHERE NAME= 'ArrayDepth' AND VALUE >50 INTERSECT SELECT DISTINCT FID FROM ACOUSTIC_METADATA WHERE NAME= 'SpeedLog' AND VALUE <20)	00:00:00.226
Q6	Project+Date+ Parameter1 (Depth) +Parameter2 (SpeedLog) +Parameter3 (Roll)	SELECT FNAME FROM FILE_DETAIL WHERE PROJECT='ATLAS_v2' AND date(date)>'2020-01-01' AND date(date)<'2020-12-01' AND FID IN (SELECT DISTINCT FID FROM ACOUSTIC_METADATA WHERE NAME= 'ArrayDepth' AND VALUE <100 INTERSECT SELECT DISTINCT FID FROM ACOUSTIC_METADATA WHERE NAME= 'ArrayDepth' AND VALUE >50 INTERSECT SELECT DISTINCT FID FROM ACOUSTIC_METADATA WHERE NAME= 'SpeedLog' AND VALUE <20 INTERSECT SELECT DISTINCT FID FROM ACOUSTIC_METADATA WHERE NAME= 'Roll' AND VALUE >90)	00:00:00.272

TABLE II: Sample use cases on querying file system based on metadata

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